

RM- Assignment 1

Phase 1: Deciding

1. Identifying Research Problem

Identifying Research the research problem by studying how the rapid growth of AI workloads (training and inference) is driving increased water consumption in data centre cooling systems, and understand why current cooling methods fail to balance thermal efficiency with environmental sustainability. The study focuses on quantifying water usage effectiveness across cooling methods and finding practical alternatives that reduce freshwater dependency without compromising system reliability.

2. Literature Review

Review existing cooling technologies (air, evaporative, liquid, immersion) and water usage metrics. Identify the gap i.e. lack of standardized comparison of cooling methods specifically for AI's high-density CPU workloads.

3. Define objectives and Scope.

Evaluate water consumption across cooling architectures used in AI data centers and compare alternatives (closed-loop liquid, immersion, recycled water) on water use, energy use and performance.

Phase 2: Planning

4. Research Design

Compare cooling types (cooling towers, indirect liquid, immersion, air cooled) using simulation or case-study data across different AI workload intensities (training vs inference).

5. Data ~~Collection~~ ~~Analysis~~ Collection.

Gather ~~Data~~ ^{data} from ESG/sustainability reports of major AI providers (Google, Microsoft, Meta, AWS) and simulation tools, standardizing metrics like WUE and PUE for comparison.

6. Data Analysis

Evaluate and compare WUE, PUE and water-to-compute ratios across cooling methods to see which best suits high-density AI loads.

7. Phase 7: Execution

7. Interpretation and Results

Identify which cooling method best balances water savings, energy efficiency and reliability. Note limitations (cost, scalability) and regional factors (water-scarce locations).

8. Report / Thesis / Paper Writing

Document methodology, results and conclusions. Present in suitable journal / conference.